

SMART INDIA HACKATHON 2026



SUKOBIN

Carriers as the sensor network

Problem Statement ID – 26002

Problem Statement Title – AI-Enabled Logistics Accessibility Intelligence Platform for the North Eastern Region

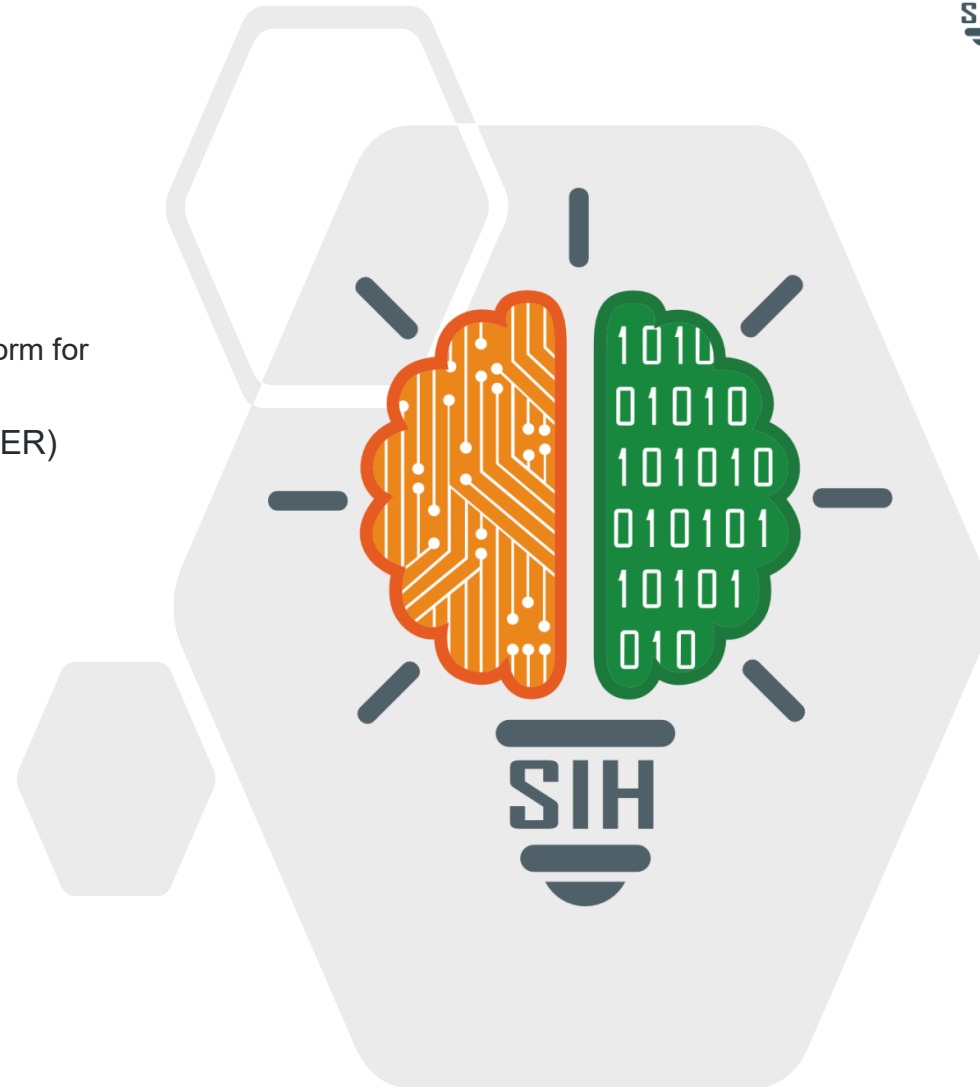
Organization – Ministry of Development of North Eastern Region (MDoNER)

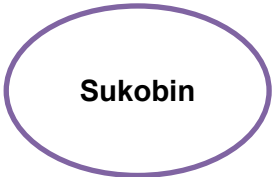
Theme – Transportation & Logistics

PS Category – Software

Team ID – <team id>

Team Name – Sukobin





EVERY VEHICLE ON THE ROAD IS A SENSOR



DETAILED EXPLANATION OF THE PROPOSED SOLUTION

Nobody drives for us. A driver declares a journey they were making anyway, and is offered only the consignments lying along that road, within capacity. Instead of costly roadside sensors, vehicles already travelling the region become mobile road probes.

Their movement is the measurement. Median speed against each road's baseline gives live accessibility; AI turns it into risk scores, 72-hour forecasts, alerts and alternate routes.

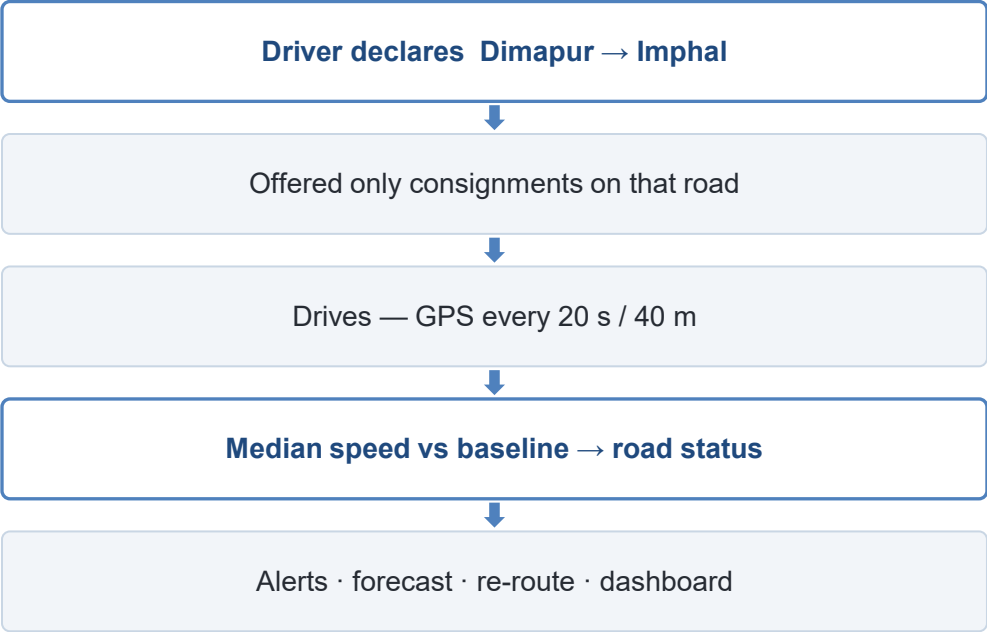
INNOVATION AND UNIQUENESS OF THE SOLUTION

Carriers ARE the sensors one journey delivers goods and reads the road	No hardware, no fleet nothing to install, power or maintain
Voice reporting, 10 languages speak it; the model classifies and reads it back	Offline-first field reports queue on the phone, sync later, never filed twice
Explainable forecasts every number decomposes into the feature that drove it	A trust ladder an unverified report can slow a road, never close one
Photo + GPS evidence every incident carries proof, not a phone call	It admits what it cannot see coverage is published, so grey never looks like broken

HOW IT ADDRESSES THE PROBLEM — EVERY CLAUSE OF THE PROBLEM STATEMENT

(a) Road & bridge accessibility 42 stretches, 3,567 km, 82 districts, live from GPS	(b) Disruption prediction 24 / 48 / 72 h closure risk on every road	(c) Alternate routes + delay 3 alternatives scored, condition-adjusted ETA	(d) GPS tracking of essentials medicines, food, produce, construction material
(e) Automated alerts blocked road, cut-off region, high-risk corridor	(f) Field reporting photo + GPS + voice, queued when offline	(g) Central dashboard district status, bottlenecks, emergency routes	(h) Multilingual + offline 10 languages, sync when the signal returns

ONE TRIP, TWO PRODUCTS



TECHNOLOGIES TO BE USED (PROGRAMMING LANGUAGES, FRAMEWORKS, HARDWARE)

SOFTWARE STACK

Mobile Kotlin + XML — 4 Android apps: customer, merchant, partner, officer

Web React · Vite · MapLibre GL dashboard

Backend Node.js · Express 5 · MongoDB 2dsphere · JWT

Cloud Render · Atlas · Cloudinary · FCM

AI / ML AND DATA

Model Logistic regression vs gradient-boosted stumps, 18 features

Trained on 109,116 road-days · AUC 0.883 · Brier 0.092

Weather Open-Meteo archive + forecast, hourly

Roads OSRM geometry and route alternates · VAHAN lookup

PLATFORM COMPONENTS

Hardware None — the driver's own phone is the sensor

Engines Route prediction · GIS monitoring · GPS tracking · alerts

Field layer Photo, GPS and voice reporting in 10 languages

Resilience Offline queue, cloud storage, secure token auth

METHODOLOGY AND PROCESS FOR IMPLEMENTATION



ALGORITHM 1 — CORRIDOR MATCHING

- 1 Coarse fetch — bounding box around the route polyline
- 2 Corridor — pickup and drop ≤ 10 km off the line, or inside the endpoint city
- 3 Detour cap — mid-route deviation ≤ 24 km, else rejected
- 4 Direction — driver $\rightarrow$ pickup $\rightarrow$ drop, projected along the line
- 5 Score, then sequence stops by position so the route never doubles back

$$\text{score} = 1.0 \times \text{fee} - 8.0 \times \text{offRouteKm} - 0.15 \times \text{ageMinutes}$$

ALGORITHM 2 — PROBE SENSING

45-minute rolling median of vehicle speed $\div$ the road's own baseline:

- ≤ 0.15 $\rightarrow$ **BLOCKED**
- ≤ 0.35 $\rightarrow$ **RESTRICTED**
- ≤ 0.60 $\rightarrow$ **SLOW**
- > 0.60 $\rightarrow$ **OPEN**

Trust rule — ≥ 4 samples from ≥ 2 distinct vehicles, else discarded. An unverified driver report is capped at RESTRICTED.

ANALYSIS OF THE FEASIBILITY OF THE IDEA

Runs on smartphones, GPS, GIS and cloud services — no roadside hardware to procure, install or power. Modular, so weather, routing and government sources can be swapped per state.

42 stretches · 3,567 km real OSRM geometry, 12 corridors, 8 states

4 apps + dashboard built and running against live data

195 / 195 tests 10 suites · 10 languages · 0 English fallbacks

BUSINESS MODEL — HOW IT SUSTAINS ITSELF

Flat platform fee ₹5 per parcel, ₹2 per shop order. The driver keeps the full delivery charge — that is what makes carrying worth it.

Government licence the accessibility dashboard sold per state or district to transport and disaster-management departments.

Merchant plans optional listing and analytics tiers for shops.

Cost side cloud only. No fleet, no vehicles, no hardware, so marginal cost per extra parcel is a payment-gateway fee.

POTENTIAL CHALLENGES AND RISKS

Remote areas lose network

Noisy or inaccurate GPS

Incorrect user reports

Uncertainty in AI predictions

Roads close mid-journey

No historical closure dataset

STRATEGIES FOR OVERCOMING THESE CHALLENGES

Field reports queue on the phone under a client ID and sync later; the server rejects duplicates, so nothing is filed twice.

Fixes worse than 120 m are dropped on the device, and a road status needs ≥ 4 samples from ≥ 2 distinct vehicles before it changes.

An unverified report is capped at RESTRICTED; only an officer's verification can close a road.

Every prediction is explainable by feature, calibration is published on screen, and current risk is kept separate from the forecast.

The route planner scores three real alternatives and reports why each rejected one failed.

Labels come from a rainfall-threshold hazard function; verified field reports override them, and the platform says so openly.

POTENTIAL IMPACT ON THE TARGET AUDIENCE

District administration

Continuous visibility of accessibility, and a lifeline corridor's closure risk 72 hours out.

Field officers

Weak points ranked with the reason each scored, and a queue of reports to verify.

Transporters and drivers

Income from a journey already being made, and route conditions before setting out.

People in remote blocks

Medicines, food, produce and construction material arrive — with an alert in the language spoken at home when they will not.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

Social

8 states, 82 districts. Alerts and voice reporting in 10 languages, four with almost no other software support.

Economic

Corridor matching turns spare capacity into income. No dedicated fleet, so cost per extra parcel is near zero.

Environmental

Fewer dedicated trips — a parcel rides in a vehicle already making the journey, not a second one.

Governance

Photo, GPS fix, timestamp and the verifying officer on every incident. An evidence trail, not a phone call.

PRODUCT DEVELOPMENT TIMELINE

Phase 1 **DONE**

Core platform

4 apps, dashboard, 118 endpoints

Phase 2 **DONE**

Intelligence layer

probe sensing, forecast model, alerts, 10 languages

Phase 3 **4 WEEKS**

Close the loop

road-ahead warnings, mid-trip re-route, delay alerts

Phase 4 **3 MONTHS**

District pilot

one corridor with a state transport department

Phase 5 **6-12 MONTHS**

Scale

multi-state rollout, government system integration

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

DATA SOURCES THE PLATFORM RUNS ON

Open-Meteo Historical Weather API

open-meteo.com/en/docs/historical-weather-api — observed hourly rainfall, snowfall and temperature; the model's training set

OSRM — Open Source Routing Machine

project-osrm.org — real road geometry and route alternates

VAHAN / Parivahan registration lookup

parivahan.gov.in — vehicle class and capacity at driver sign-up

MapLibre GL JS + Esri Dark Gray Canvas

maplibre.org — the GIS layer of the control dashboard

PROBLEM AND DOMAIN RESEARCH

MDoNER — problem statement 26002

mdoner.gov.in — NER connectivity gaps and infrastructure priorities

NHIDCL / MoRTH national highway network

nhidcl.com · morth.nic.in — corridor alignments and NH numbering

IMD rainfall climatology for the North East

mausam.imd.gov.in — monsoon windows and the rain thresholds used

GSI landslide susceptibility mapping

gsi.gov.in — which stretches are treated as landslide-prone

METHOD — HOW THE MODEL WAS BUILT AND JUDGED

Date-based evaluation — trained on 109,116 road-days across 42 stretches x 877 days of observed weather, with everything after 2026-03-01 held out. AUC 0.883, Brier 0.092.

Stated openly: no public register of past NER road closures exists, so historical labels are drawn from a rainfall-threshold hazard function calibrated on IMD and GSI data. Verified field reports override every drawn label.